

MTH304 - Midsem 2

All questions are worth 4 points

1. In S_5 , consider the permutations

$$\sigma := (12345) \quad \text{and} \quad \tau = (15432)$$

Find an element $x \in S_5$ such that

$$x\sigma x^{-1} = \tau$$

2. Write the set of conjugacy classes of elements in S_4 .
3. Let $\mathbb{C}^\times := \mathbb{C} \setminus \{0\}$. Consider the subgroup $G \subset GL_2(\mathbb{R})$ defined by

$$\left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \mid a, b \in \mathbb{R}, a^2 + b^2 > 0 \right\}$$

Consider $\phi : \mathbb{C}^\times \rightarrow G$ given by $a + ib \mapsto \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Show that ϕ defines an isomorphism of groups.

4. Let $S^1 := \{x \in \mathbb{C} \mid |x| = 1\}$ denote the group of complex numbers of modulus 1 under multiplication. Define a map $\phi : \mathbb{R} \rightarrow S^1$ such that the quotient group $\mathbb{R}/\mathbb{Z}$ becomes isomorphic to S^1 . Let $(\mathbb{R}_{>0}, \times)$ denote the group of positive reals under multiplication. Define a map $\Phi : \mathbb{C}^\times \rightarrow \mathbb{R}_{>0}$ such that the quotient group $\mathbb{C}^\times/S^1$ becomes isomorphic to $(\mathbb{R}_{>0}, \times)$ and prove it.
5. Let G be a group with centre Z . For $g \in G$, let $i_g : G \rightarrow G$ denote the inner automorphism of G given by $h \mapsto ghg^{-1}$. Let

$$\text{Inn}(G) = \{i_g \mid g \in G\}$$

denote the subgroup of $\text{Aut}(G)$ consisting of inner automorphisms of G . Show that the map $\phi : G \rightarrow \text{Inn}(G)$ given by $\phi(g) = i_g$ is a surjective group homomorphism. Show that

$$G/Z \cong \text{Inn}(G)$$